Homework 4

Statistical Modeling and Learning

due: **Sunday, October 25, 2026, 5:00p**

Instructions: Please submit solutions on Populi as a professionally compiled PDF (for example, from a knitted R Markdown file or Jupyter notebook). For handwritten work, please scan your solution pages and include them in the final PDF. Homeworks can be completed individually or in groups of up to three people. Names of all group members must be included on the write-ups.

Problem 1: Logistic Regression for a Marketing Campaign. Download `bank-full.csv` and `bank-names.txt` from the course website. The data come from a Portuguese bank's direct marketing campaigns (phone calls) to promote a term deposit. The outcome of interest is whether the client subscribed to the term deposit (`y`: "yes" or "no").

Key variables (see `bank-names.txt` for full descriptions):

- `age`: client age (numeric)
- `balance`: average yearly balance in euros (numeric)
- `previous`: number of contacts performed before this campaign for this client (numeric)
- `poutcome`: outcome of the previous marketing campaign (categorical: "unknown", "other", "failure", "success")
- `job`: type of job (categorical: e.g., "management", "blue-collar", "retired", "student", etc.)

Use only these variables (and `y`) in the problem below.

a. What fraction of clients in the dataset subscribed to the term deposit? Create an exploratory plot showing how subscription rate varies by one categorical predictor (e.g., `job` or `poutcome`). What do you observe?

b. Fit a logistic regression model predicting subscription (`y`) from `balance`, `age`, and `previous`. Write out the fitted model. Interpret the coefficient on `balance` in terms of odds: for every 1000 euro increase in yearly balance, how do the odds of subscribing change?

c. In the model from part (b), interpret the coefficient on `previous`. Does having been contacted in a previous campaign appear to be associated with higher or lower probability of subscribing in this campaign?

d. Add `poutcome` (treat as categorical; choose a reference category) to the model. Is previous campaign success (`poutcome` = "success") associated with higher probability of subscribing? Interpret the estimated coefficient for previous success in terms of odds.

e. Using your model from part (d), what is the predicted probability of subscribing for a client with balance = 2000 euros, age = 45, previous = 2, and `poutcome` = "success"? How does this compare to a client with the same balance, age, and previous but `poutcome` = "failure"?

f. In one or two sentences, how might the bank use a model like this to decide whom to contact in a future campaign?

Problem 2: Logistic Regression for Credit Card Default. Download the file `Default.csv` from the course website. This is the data set used throughout ISL Chapter 4: 10,000 credit card customers, with `default` ("Yes" or "No": did the customer default on their payment?), `student` ("Yes" or "No"), `balance` (average monthly credit card balance, in dollars), and `income` (annual income, in dollars). Read the data with `read.csv(..., stringsAsFactors = TRUE)`.

a. What fraction of customers defaulted? Make a plot comparing `balance` for defaulters and non-defaulters, and compute the default rate separately for students and non-students. What do you observe?

b. Fit a logistic regression of `default` on `student` alone. Interpret the coefficient on `student` in terms of odds. Based on this model alone, are students more or less likely to default?

c. Now fit `default` on `balance` and `student`. Interpret the coefficient on `balance` in terms of odds for a \$100 increase in balance. What happened to the sign of the `student` coefficient compared with part (b)? Compute the average balance for students and non-students, and use it to explain the change in two or three sentences. (Hint: what does "holding balance constant" mean here?)

d. For the model in part (c), plot the fitted probability of default against `balance`, with one curve for students and one for non-students. Then compute the predicted probability of default for a student and a non-student, each with a balance of \$1,500.

e. Add an interaction between `balance` and `student`. Write out the fitted model for students and for non-students separately. Is there evidence that the effect of balance differs between students and non-students? Use the z test for the interaction coefficient to answer.

f. For the model in part (c), test the overall null hypothesis that neither `balance` nor `student` has any predictive power, using a likelihood ratio test. Report the test statistic (the drop in deviance from the intercept-only model), its degrees of freedom, and the p-value.

g. Using the model in part (c), classify each customer as a predicted defaulter if the fitted probability exceeds 0.5. Make the confusion matrix and report the accuracy, the true positive rate (the share of actual defaulters the model catches), and the false positive rate. Compare the accuracy with a "model" that predicts no one ever defaults. Then repeat with a threshold of 0.2. If you were the credit card company, which threshold would you use, and why?

Problem 3: Should the Developer Build Green? Download the file `greenbuildings.csv` from the course website. Each row is a commercial office building in the United States. About 680 of them carry a green certification (LEED or Energy Star; `green_rating` = 1). Each building was matched with a "cluster" of nearby non-green buildings, and `cluster_rent` records the average rent per square foot in that local market. The outcome of interest is `Rent`, in dollars per square foot per year. Other variables include building `size` (square feet), `stories`, `age` (years), `renovated`, quality class (`class_a`, `class_b`; the rest are class C), `leasing_rate` (percent of space occupied), `net` (1 if tenants pay their own utilities), `amenities` (1 if on-site amenities such as a gym or restaurant), regional employment growth (`empl_gr`), and local climate and utility prices (`cd_total_07`, `hd_total07`, `total_dd_07`, `Precipitation`, `Gas_Costs`, `Electricity_Costs`). (Seventy-four rows are missing `empl_gr`; say how you handle them.)

The decision. A developer is planning a new 15-story, 250,000 square foot mixed-use office building. Baseline construction costs are $100 million, and green certification would add a 5% premium ($5 million). The developer's in-house analyst argues:

> "I compared the median rent of green buildings with that of non-green buildings: $27.60 versus $25.00 per square foot. That is a $2.60 premium. Times 250,000 square feet, that is $650,000 of extra revenue per year, so we recoup the $5 million in under eight years and then it is pure profit. We should build green."

Your task. Write a short memo to the developer (about one to two pages, plus an appendix with your code and supporting output) that answers two questions. Is the analyst's $2.60 a credible estimate of what going green would be worth for *this* building? And should the developer build green? There is no single right answer; you will be graded on whether your recommendation is supported by the data and on how clearly you explain it. Your memo must:

- Explain what, if anything, is wrong with the analyst's comparison. Are green and non-green buildings otherwise alike? Support your answer with numbers or a plot.
- Estimate the green premium with a multiple regression that holds other building characteristics constant. You decide which predictors belong in the model, but use at least one of the model selection tools from class (adjusted R^2, AIC, best subsets, stepwise selection, or a train/test split) to choose or defend it, and say what each tool told you. Do not put `LEED` or `Energystar` in alongside `green_rating` (they are pieces of the same variable), and leave out the ID columns.
- Report your estimated premium with a 95% confidence interval, and show how sensitive it is to the choice of model. Does the estimate change much across reasonable models?
- Remember that the developer cares about revenue, not just the posted rent: a building that charges more but sits half empty is not a good investment. Consider whether going green also affects how much of the building gets leased (`leasing_rate`), and what that means for revenue.
- Translate your estimate into dollars for this building and give a payback period, with a range that reflects your uncertainty. End with a clear recommendation and the one or two caveats the developer should keep in mind.

Problem 4: Penalized Regression for House Prices. Download the file `saratoga_houses.csv` from the course website. Each row is a house sold in Saratoga County, New York. The outcome is `price`. There are 15 candidate predictors, six of them categorical. Read the data with `read.csv(..., stringsAsFactors = TRUE)` so that the categorical variables become factors.

a. Fit the main-effects model `lm(price ~ ., data = houses)` and the model with every main effect and every pairwise interaction, `lm(price ~ .^2, data = houses)`. How many coefficients does the interaction model have? Some of them are `NA`. Why? (Hint: think about combinations of categories that never occur together in the data.) Compare R^2 and adjusted R^2 for the two models. Based on these in-sample numbers alone, which model looks better?

b. Split the data 70/30 into training and test sets, running `set.seed(42)` immediately before you `sample()` the training rows. On the *training set only*, fit (i) all main effects, (ii) backward stepwise starting from (i), and (iii) all pairwise interactions. Make a table of the number of estimated coefficients, training RMSE, and test RMSE for each model. (R may warn that predictions come from a rank-deficient fit; that warning is itself a symptom worth mentioning.) Which model predicts best on the test set? Which overfits worst?

c. Instead of throwing the interactions away, we can keep all of them and shrink their coefficients. Build the predictor matrix and response:

```
X <- model.matrix(price ~ .^2, data = houses)[, -1]  # no intercept column
y <- houses$price
```

and use exactly the same training and test rows as in part (b). Fit ridge regression on the training set with `glmnet(X.train, y.train, alpha = 0)` and plot the coefficient path with `plot(..., xvar = "lambda")`. Then run `set.seed(42)` and choose λ by 10-fold cross-validation with `cv.glmnet(X.train, y.train, alpha = 0)`. Plot the CV curve and report $\lambda_{\min}$ and $\lambda_{1\text{se}}$. Do any coefficients go exactly to zero? In one sentence, why does it matter that `glmnet` puts the predictors on a common scale before penalizing?

d. Repeat part (c) for the LASSO (`alpha = 1`), again running `set.seed(42)` before `cv.glmnet`. Report $\lambda_{\min}$, $\lambda_{1\text{se}}$, and the number of nonzero coefficients at each. List the variables the LASSO keeps at $\lambda_{1\text{se}}$. How many are main effects, and how many are interactions? Pick two of the surviving interactions and explain in plain language what each one says about house prices.

e. Compute the test-set RMSE for ridge and the LASSO, each at both $\lambda_{\min}$ and $\lambda_{1\text{se}}$, and add them to your table from part (b). How does the LASSO with 168 candidate predictors compare with least squares on the same 168 predictors? With the main-effects model? Explain the comparison in terms of the bias–variance tradeoff.

f. The one-standard-error rule picks a simpler model than $\lambda_{\min}$, at some cost in test RMSE here. In two or three sentences, argue for which of the two you would hand to a client who wants to understand what drives house prices, and which you would hand to a client who only wants the most accurate price predictions.